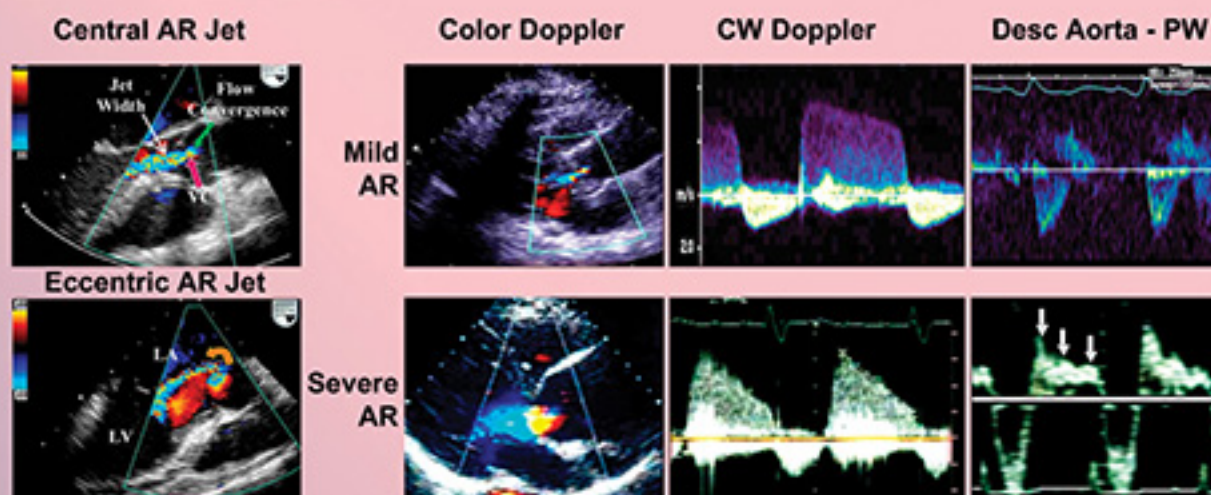


Aortic Valve



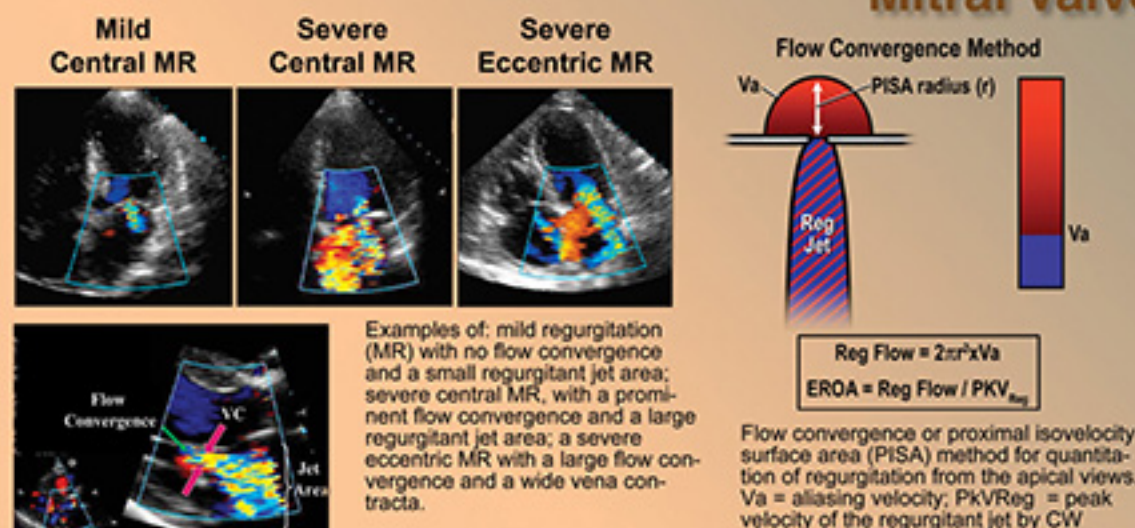
Central and eccentric aortic regurgitation (AR) jets. VC = vena contracta; LA = left atrium; LV = left ventricle. Color Doppler, continuous wave, and pulsed wave (PW) Doppler recording of flow in the descending aorta in mild and severe aortic regurgitation (AR). Arrows: holodiastolic flow reversal in the descending (desc) aorta.

Grading of Aortic Regurgitation Severity

	Mild	Moderate	Severe
Specific Signs for AR severity	- Central Jet, width <25% of LVOT - Vena contracta <0.3 cm¹ - No or brief early diastolic flow reversal in descending aorta	Signs of AR>mild present but no criteria for severe AR	- Central Jet, width ≥65% of LVOT - Vena contracta > 0.6 cm
Supportive Signs	- Pressure half-time > 500 ms - Normal LV size²	Intermediate values	- Pressure half-time <200 ms - Holodiastolic aortic flow reversal in descending aorta - Moderate or greater LV enlargement³
Quantitative Parameters			
RVol, ml/beat	< 30	30-44	45-59
RF, %	< 30	30-39	40-49
EROA, cm ²	< 0.10	0.10-0.19	0.20-0.29

¹ At a Nyquist limit of 50-60 cm/s.
² LV size applied only to chronic lesions.
³ In the absence of other etiologies of LV dilatation.
 AR = aortic regurgitation; EROA = effective regurgitant orifice area; LV = left ventricle; LVOT = left ventricular outflow tract; RVol = regurgitant volume; RF = regurgitant fraction.

Mitral Valve



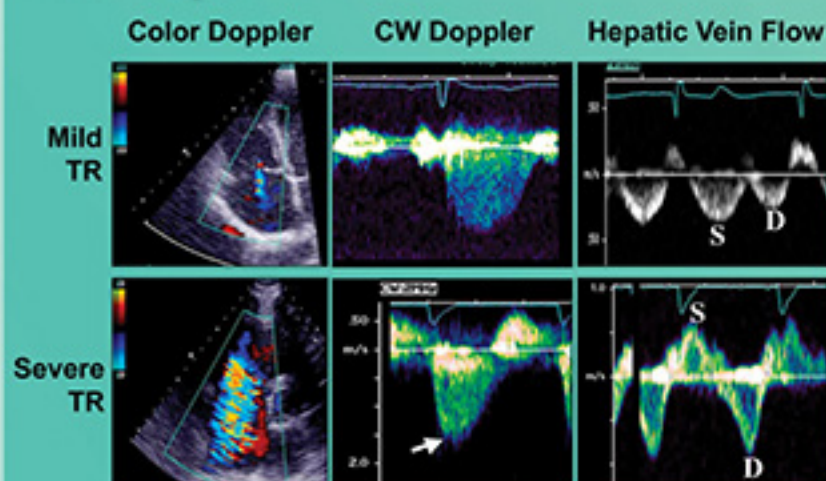
Mitral regurgitation jet depicting its 3 components: flow convergence, vena contracta (VC), and jet area in the left atrium.

Grading Mitral Regurgitation Severity

	Mild	Moderate	Severe
Specific Signs of Severity	- Small central jet <4 cm² or < 20% of LA area. - Vena contracta width < 0.3 cm - No or minimal flow convergence¹	Signs of MR>mild present but no criteria for severe MR	- Vena contracta width ≥ 0.7 cm with large central MR Jet (area > 40% of LA) or with a wall-impinging jet of any size, swirling in LA - Large flow convergence¹ - Systolic reversal in pulmonary veins - Prominent flail MV leaflet or ruptured papillary muscle
Supportive Signs	- Systolic dominant flow in pulmonary veins - A-wave dominant mitral inflow² - Soft density, parabolic CW Doppler MR signal - Normal LV size³	Intermediate signs/findings	- Dense, triangular CW Doppler MR jet - E-wave dominant mitral inflow (E > 1.2 m/s)² - Enlarged LV and LA size, particularly when normal LV function is present
Quantitative Parameters			
RVol (ml/beat)	<30	30-44	45-59
RF (%)	<30	30-39	40-49
EROA (cm ²)	<0.20	0.20-0.29	0.30-0.39

Color Nyquist limit of 50-60 cm/s.
¹ Minimal and large flow convergence defined as a flow convergence radius < 0.4 cm and ≥ 0.9 cm for central jets, respectively, with a baseline shift at a Nyquist of 40 cm/s.
² Usually above 50 years of age or in conditions of impaired relaxation, in the absence of mitral stenosis or other causes of elevated LA pressure.
³ LV size applied only to chronic lesions.
 CW = continuous wave; EROA = effective regurgitant orifice area; LA = left atrium; LV = left ventricle; MV = mitral valve; MR = mitral regurgitation; RVol = regurgitant volume; RF = regurgitant fraction.

Tricuspid Valve



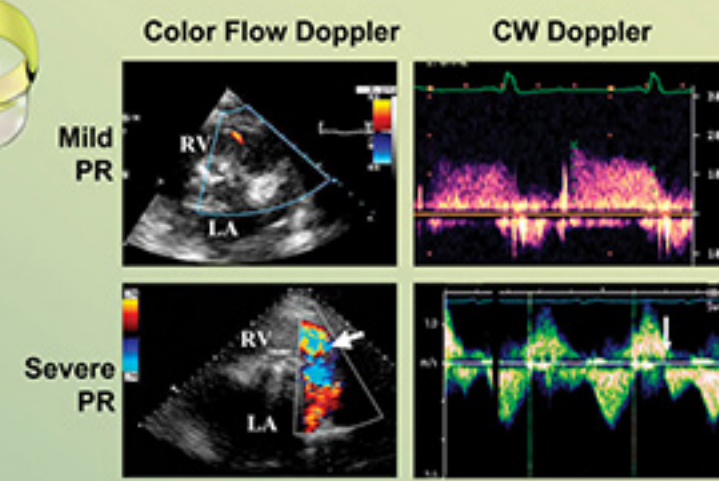
Jet recordings by color Doppler, continuous wave Doppler, and hepatic vein flow by pulsed Doppler in mild and severe tricuspid regurgitation (TR). Systole = S; Diastole = D.

Grading Tricuspid Regurgitation Severity

Parameter	Mild	Moderate	Severe
Tricuspid valve	Usually normal	Normal or abnormal	Abnormal/Flail leaflet/Poor coaptation
RV/RA/IVC size	Normal ¹	Normal or dilated	Usually dilated ²
Jet area- central jets (cm ²) ³	< 5	5-10	>10
VC width (cm)	Not defined	Not defined, but <0.7	>0.7
PISA radius (cm) ⁴	<0.5	0.6 - 0.9	>0.9
Jet density and contour -CW	Soft and parabolic	Dense, variable contour	Dense, triangular with early peaking
Hepatic vein flow ⁵	Systolic dominance	Systolic blunting	Systolic reversal

¹ Unless there are other reasons for RA or RV dilation; ² Exception: acute TR.
³ At a Nyquist limit of 50-60 cm/s.
⁴ Baseline shift with Nyquist limit of 28 cm/s.
⁵ Other conditions may cause systolic blunting (eg. atrial fibrillation, elevated RA pressure).
 CW = Continuous wave Doppler; IVC = inferior vena cava; RA = right atrium; RV = right ventricle; VC = vena contracta width.

Pulmonic Valve



Color flow and continuous wave Doppler recording in mild and severe pulmonary regurgitation (PR). Arrow: early termination of PR flow; LA = left atrium; RV = right ventricle.

Grading Pulmonary Regurgitation Severity

Parameter	Mild	Moderate	Severe
Pulmonic valve	Normal	Normal or abnormal	Abnormal
RV size	Normal ¹	Normal or dilated	Dilated ²
Jet size by color Doppler	Thin (usually <10 mm in length) with a narrow origin	Intermediate	Usually large, with a wide origin; May be brief in duration
Jet density and deceleration rate -CW ³	Soft; Slow deceleration	Dense; variable deceleration	Dense; steep deceleration, early termination of diastolic flow
Pulmonic systolic flow compared to systemic flow -PW	Slightly increased	Intermediate	Greatly increased

¹ Unless there are other reasons for RV enlargement. ² Exception: acute PR.
³ Steep deceleration is not specific for severe PR.
 CW = Continuous wave Doppler; PR = pulmonic regurgitation; PW = pulsed wave Doppler; RV = right ventricle.

Poster ordering information and full text of ASE guideline documents available at: www.asecho.org

Adapted from: Zoghbi WA, Enriquez-Sarano M, Foster E, Grayburn PA, Kraft CD, Levine RA, Nihoyannopoulos P, Otto CM, Quinones MA, Rakowski H, Stewart WJ, Waggoner A, Weissman NJ.

Recommendations for Evaluation of The Severity of Native Valvular Regurgitation with Two-Dimensional and Doppler Echocardiography. J Am Soc Echocardiogr. 2003 Jul;16(7):777-802.

Recommendations endorsed by the American College of Cardiology, the American Heart Association, and the European Society of Cardiology Working Group on Echocardiography.

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